LSTM

W zadaniach NLP (Natural Language Processing), takich jak:

- analiza sentymentu,
- tłumaczenie maszynowe,
- generowanie tekstu,
- predykcja kolejnego słowa, potrzebujemy modeli, które rozumieją sekwencję słów, a nie tylko pojedyncze słowa.

Tu właśnie używamy **LSTM** — sieci, która potrafi pamiętać kontekst w czasie (np. zależności gramatyczne lub semantyczne).

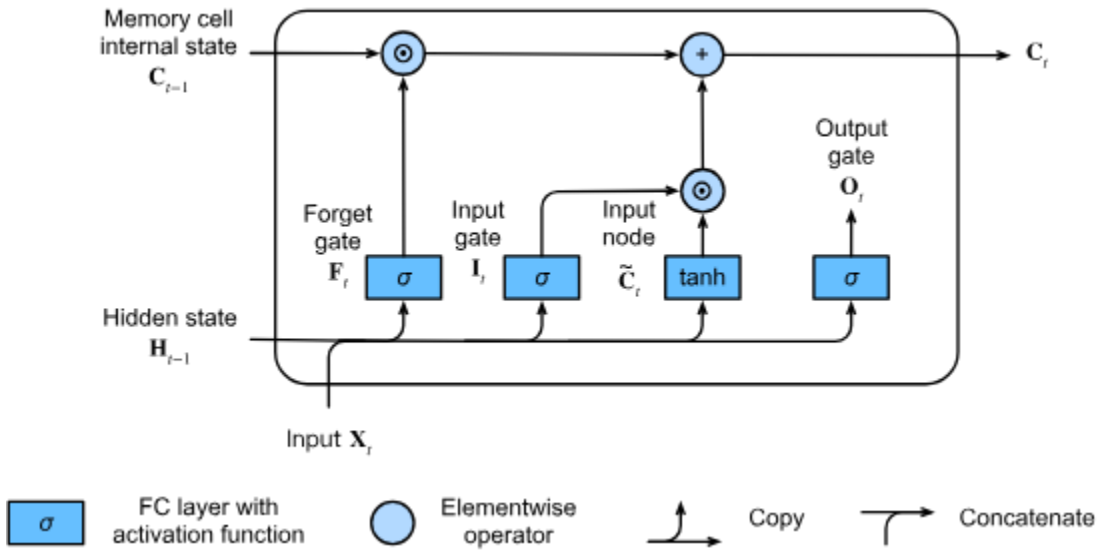
Kiedy używać LSTM?

- gdy sekwencje mają od umiarkowanej do dużej długości i **zależności czasowe są ważne**
- gdy mamy mało danych i nie chcemy używać cięższych modeli (np. transformery - to już niedługo :)), albo chcemy modelu, który działa dobrze na mniejszych zbiorach
- do zadań sekwencyjnych

Kiedy nie używać?

- gdy mamy ogromne zbiory i potrzebujemy najlepszych wyników — transformery (np. BERT) często biją LSTM
- gdy zadanie wymaga globalnych zależności i kontekstów, które lepiej łapią mechanizmy self-attention

Jak zbudowany jest LSTM?



Podstawowy blok LSTM składa się z: stanu komórkowego (cell state) i wewnętrznych bramek:

- **forget gate** (bramka zapominania) — decyduje, które informacje z poprzedniej komórki zignorować
- **input gate** (bramka wejścia) — decyduje, które nowe informacje dodać

- **output gate** (bramka wyjścia) — decyduje, jaka część stanu komórkowego stanie się wyjściem

Przykład - przewidywanie kolejnego słowa w zdaniu

In [1]:
```
!pip install flair
```

```
Collecting flair
  Downloading flair-0.15.1-py3-none-any.whl.metadata (12 kB)
Collecting boto3 ⩾ 1.20.27 (from flair)
  Downloading boto3-1.43.117-py3-none-any.whl.metadata (6.6 kB)
Collecting conllu<5.0.0,⩾4.0 (from flair)
  Downloading conllu-4.5.3-py2.py3-none-any.whl.metadata (19 kB)
Collecting deprecated ⩾ 1.2.13 (from flair)
  Downloading deprecated-3.0.0-py3-none-any.whl.metadata (6.5 kB)
Collecting ftfy ⩾ 6.1.0 (from flair)
  Downloading ftfy-6.3.1-py3-none-any.whl.metadata (7.3 kB)
Requirement already satisfied: gdown ⩾ 4.4.0 in
/usr/local/lib/python3.13/dist-packages (from flair) (5.2.2)
Requirement already satisfied: huggingface-hub ⩾ 0.10.0 in
/usr/local/lib/python3.13/dist-packages (from flair) (1.33.0)
Collecting langdetect ⩾ 1.0.9 (from flair)
  Downloading langdetect-1.0.9.tar.gz (981 kB)
     ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 981.5/981.5 kB 19.0 MB/s eta
0:00:00
  Preparing metadata (setup.py) ... done
Requirement already satisfied: lxml ⩾ 4.8.0 in
/usr/local/lib/python3.13/dist-packages (from flair) (6.1.3)
Requirement already satisfied: matplotlib ⩾ 2.2.3 in
/usr/local/lib/python3.13/dist-packages (from flair) (3.10.0)
Requirement already satisfied: more-itertools ⩾ 8.13.0 in
/usr/local/lib/python3.13/dist-packages (from flair) (10.8.0)
Collecting mpld3 ⩾ 0.3 (from flair)
  Downloading mpld3-0.5.12-py3-none-any.whl.metadata (5.3 kB)
Collecting pptree ⩾ 3.1 (from flair)
  Downloading pptree-3.1.tar.gz (3.0 kB)
  Preparing metadata (setup.py) ... done
Requirement already satisfied: python-dateutil ⩾ 2.8.2 in
/usr/local/lib/python3.13/dist-packages (from flair) (2.9.0.post0)
Collecting pytorch-revgrad ⩾ 0.2.0 (from flair)
  Downloading pytorch_revgrad-0.2.0-py3-none-any.whl.metadata (1.7 kB)
Requirement already satisfied: regex ⩾ 2022.1.18 in
/usr/local/lib/python3.13/dist-packages (from flair) (2025.11.3)
Requirement already satisfied: scikit-learn ⩾ 1.0.2 in
/usr/local/lib/python3.13/dist-packages (from flair) (1.6.1)
Collecting segtok ⩾ 1.5.11 (from flair)
  Downloading segtok-1.5.11-py3-none-any.whl.metadata (9.0 kB)
Collecting sqlitedict ⩾ 2.0.0 (from flair)
  Downloading sqlitedict-2.1.0.tar.gz (21 kB)
  Preparing metadata (setup.py) ... done
Requirement already satisfied: tabulate ⩾ 0.8.10 in
/usr/local/lib/python3.13/dist-packages (from flair) (0.9.0)
Requirement already satisfied: torch ⩾ 1.13.1 in
/usr/local/lib/python3.13/dist-packages (from flair) (2.11.0+cu130)
Requirement already satisfied: tqdm ⩾ 4.63.0 in
/usr/local/lib/python3.13/dist-packages (from flair) (4.67.3)
```

```
Collecting transformer-smaller-training-vocab≥0.2.3 (from flair)
  Downloading transformer_smaller_training_vocab-0.4.2-py3-none-
any.whl.metadata (7.9 kB)
Collecting transformers<5.0.0,≥4.25.0 (from transformers[sentencepiece]
<5.0.0,≥4.25.0→flair)
  Downloading transformers-4.57.6-py3-none-any.whl.metadata (43 kB)
     ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 44.0/44.0 kB 4.6 MB/s eta
0:00:00
Collecting wikipedia-api≥0.5.7 (from flair)
  Downloading wikipedia_api-0.16.0-py3-none-any.whl.metadata (44 kB)
     ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 44.7/44.7 kB 4.8 MB/s eta
0:00:00
Collecting bioc<3.0.0,≥2.0.0 (from flair)
  Downloading bioc-2.1-py3-none-any.whl.metadata (4.6 kB)
Collecting jsonlines≥1.2.0 (from bioc<3.0.0,≥2.0.0→flair)
  Downloading jsonlines-4.0.0-py3-none-any.whl.metadata (1.6 kB)
Collecting intervaltree (from bioc<3.0.0,≥2.0.0→flair)
  Downloading intervaltree-3.2.1-py2.py3-none-any.whl.metadata (12 kB)
Collecting docopt (from bioc<3.0.0,≥2.0.0→flair)
  Downloading docopt-0.6.2.tar.gz (25 kB)
  Preparing metadata (setup.py) ... done
Collecting botocore<1.44.0,≥1.43.117 (from boto3≥1.20.27→flair)
  Downloading botocore-1.43.117-py3-none-any.whl.metadata (5.6 kB)
Collecting jmespath<2.0.0,≥0.7.1 (from boto3≥1.20.27→flair)
  Downloading jmespath-1.1.0-py3-none-any.whl.metadata (7.6 kB)
Collecting s3transfer<0.20.0,≥0.19.0 (from boto3≥1.20.27→flair)
  Downloading s3transfer-0.19.3-py3-none-any.whl.metadata (1.7 kB)
Requirement already satisfied: wrapt<3,≥1.16 in
/usr/local/lib/python3.13/dist-packages (from deprecated≥1.2.13→flair)
(2.5.1)
Requirement already satisfied: wcwidth in /usr/local/lib/python3.13/dist-
packages (from ftfy≥6.1.0→flair) (0.9.2)
Requirement already satisfied: beautifulsoup4 in
/usr/local/lib/python3.13/dist-packages (from gdown≥4.4.0→flair) (4.13.5)
Requirement already satisfied: filelock in /usr/local/lib/python3.13/dist-
packages (from gdown≥4.4.0→flair) (4.2.1)
Requirement already satisfied: requests[socks] in
/usr/local/lib/python3.13/dist-packages (from gdown≥4.4.0→flair) (2.32.4)
Requirement already satisfied: click<9.0.0,≥8.4.2 in
/usr/local/lib/python3.13/dist-packages (from huggingface-hub≥0.10.0
→flair) (8.5.0)
Requirement already satisfied: fsspec≥2023.5.0 in
/usr/local/lib/python3.13/dist-packages (from huggingface-hub≥0.10.0
→flair) (2025.12.0)
Requirement already satisfied: hf-xet<2.0.0,≥1.6.0 in
/usr/local/lib/python3.13/dist-packages (from huggingface-hub≥0.10.0
→flair) (1.7.0)
Requirement already satisfied: httpx<1,≥0.23.0 in
/usr/local/lib/python3.13/dist-packages (from huggingface-hub≥0.10.0
→flair) (0.28.1)
Requirement already satisfied: packaging≥20.9 in
/usr/local/lib/python3.13/dist-packages (from huggingface-hub≥0.10.0
→flair) (26.3)
Requirement already satisfied: pyyaml≥5.1 in
```

```
 /usr/local/lib/python3.13/dist-packages (from huggingface-hub⩾0.10.0
→flair) (6.0.3)
 Requirement already satisfied: typing-extensions⩾4.1.0 in
 /usr/local/lib/python3.13/dist-packages (from huggingface-hub⩾0.10.0
→flair) (4.16.0)
 Requirement already satisfied: six in /usr/local/lib/python3.13/dist-
 packages (from langdetect⩾1.0.9→flair) (1.17.0)
 Requirement already satisfied: contourpy⩾1.0.1 in
 /usr/local/lib/python3.13/dist-packages (from matplotlib⩾2.2.3→flair)
 (1.4.0)
 Requirement already satisfied: cycler⩾0.10 in
 /usr/local/lib/python3.13/dist-packages (from matplotlib⩾2.2.3→flair)
 (0.12.1)
 Requirement already satisfied: fonttools⩾4.22.0 in
 /usr/local/lib/python3.13/dist-packages (from matplotlib⩾2.2.3→flair)
 (4.67.0)
 Requirement already satisfied: kiwisolver⩾1.3.1 in
 /usr/local/lib/python3.13/dist-packages (from matplotlib⩾2.2.3→flair)
 (1.5.1)
 Requirement already satisfied: numpy⩾1.23 in
 /usr/local/lib/python3.13/dist-packages (from matplotlib⩾2.2.3→flair)
 (2.1.3)
 Requirement already satisfied: pillow⩾8 in /usr/local/lib/python3.13/dist-
 packages (from matplotlib⩾2.2.3→flair) (11.3.0)
 Requirement already satisfied: pyparsing⩾2.3.1 in
 /usr/local/lib/python3.13/dist-packages (from matplotlib⩾2.2.3→flair)
 (3.3.3)
 Requirement already satisfied: jinja2 in /usr/local/lib/python3.13/dist-
 packages (from mpld3⩾0.3→flair) (3.1.6)
 Requirement already satisfied: scipy⩾1.6.0 in
 /usr/local/lib/python3.13/dist-packages (from scikit-learn⩾1.0.2→flair)
 (1.16.3)
 Requirement already satisfied: joblib⩾1.2.0 in
 /usr/local/lib/python3.13/dist-packages (from scikit-learn⩾1.0.2→flair)
 (1.6.0)
 Requirement already satisfied: threadpoolctl⩾3.1.0 in
 /usr/local/lib/python3.13/dist-packages (from scikit-learn⩾1.0.2→flair)
 (3.7.0)
 Requirement already satisfied: setuptools<82 in
 /usr/local/lib/python3.13/dist-packages (from torch⩾1.13.1→flair)
 (80.10.2)
 Requirement already satisfied: sympy⩾1.13.3 in
 /usr/local/lib/python3.13/dist-packages (from torch⩾1.13.1→flair) (1.14.0)
 Requirement already satisfied: networkx⩾2.5.1 in
 /usr/local/lib/python3.13/dist-packages (from torch⩾1.13.1→flair) (3.7)
 Requirement already satisfied: cuda-toolkit==13.0.2 in
 /usr/local/lib/python3.13/dist-packages (from cuda-
 toolkit[cublas,cudart,cufft,cufile,cupti,curand,cusolver,cusparse,nvjitlink,
 nvrtc,nvtx]==13.0.2; platform_system == "Linux"→torch⩾1.13.1→flair)
 (13.0.2)
 Requirement already satisfied: cuda-bindings<14,⩾13.0.3 in
 /usr/local/lib/python3.13/dist-packages (from torch⩾1.13.1→flair) (13.0.3)
 Requirement already satisfied: nvidia-cudnn-cu13==9.19.0.56 in
 /usr/local/lib/python3.13/dist-packages (from torch⩾1.13.1→flair)
```

```
(9.19.0.56)
Requirement already satisfied: nvidia-cusparselt-cu13==0.8.0 in
/usr/local/lib/python3.13/dist-packages (from torch>=1.13.1->flair) (0.8.0)
Requirement already satisfied: nvidia-nccl-cu13==2.28.9 in
/usr/local/lib/python3.13/dist-packages (from torch>=1.13.1->flair) (2.28.9)
Requirement already satisfied: nvidia-nvshmem-cu13==3.4.5 in
/usr/local/lib/python3.13/dist-packages (from torch>=1.13.1->flair) (3.4.5)
Requirement already satisfied: triton==3.6.0 in
/usr/local/lib/python3.13/dist-packages (from torch>=1.13.1->flair) (3.6.0)
Requirement already satisfied: nvidia-cublas==13.1.0.3.* in
/usr/local/lib/python3.13/dist-packages (from cuda-
toolkit[cublas,cudart,cufft,cufile,cupti,curand,cusolver,cusparse,nvjitlink,
nvrtc,nvtx]==13.0.2; platform_system == "Linux"->torch>=1.13.1->flair)
(13.1.0.3)
Requirement already satisfied: nvidia-cuda-runtime==13.0.96.* in
/usr/local/lib/python3.13/dist-packages (from cuda-
toolkit[cublas,cudart,cufft,cufile,cupti,curand,cusolver,cusparse,nvjitlink,
nvrtc,nvtx]==13.0.2; platform_system == "Linux"->torch>=1.13.1->flair)
(13.0.96)
Requirement already satisfied: nvidia-cufft==12.0.0.61.* in
/usr/local/lib/python3.13/dist-packages (from cuda-
toolkit[cublas,cudart,cufft,cufile,cupti,curand,cusolver,cusparse,nvjitlink,
nvrtc,nvtx]==13.0.2; platform_system == "Linux"->torch>=1.13.1->flair)
(12.0.0.61)
Requirement already satisfied: nvidia-cufile==1.15.1.6.* in
/usr/local/lib/python3.13/dist-packages (from cuda-
toolkit[cublas,cudart,cufft,cufile,cupti,curand,cusolver,cusparse,nvjitlink,
nvrtc,nvtx]==13.0.2; platform_system == "Linux"->torch>=1.13.1->flair)
(1.15.1.6)
Requirement already satisfied: nvidia-cuda-cupti==13.0.85.* in
/usr/local/lib/python3.13/dist-packages (from cuda-
toolkit[cublas,cudart,cufft,cufile,cupti,curand,cusolver,cusparse,nvjitlink,
nvrtc,nvtx]==13.0.2; platform_system == "Linux"->torch>=1.13.1->flair)
(13.0.85)
Requirement already satisfied: nvidia-curand==10.4.0.35.* in
/usr/local/lib/python3.13/dist-packages (from cuda-
toolkit[cublas,cudart,cufft,cufile,cupti,curand,cusolver,cusparse,nvjitlink,
nvrtc,nvtx]==13.0.2; platform_system == "Linux"->torch>=1.13.1->flair)
(10.4.0.35)
Requirement already satisfied: nvidia-cusolver==12.0.4.66.* in
/usr/local/lib/python3.13/dist-packages (from cuda-
toolkit[cublas,cudart,cufft,cufile,cupti,curand,cusolver,cusparse,nvjitlink,
nvrtc,nvtx]==13.0.2; platform_system == "Linux"->torch>=1.13.1->flair)
(12.0.4.66)
Requirement already satisfied: nvidia-cusparse==12.6.3.3.* in
/usr/local/lib/python3.13/dist-packages (from cuda-
toolkit[cublas,cudart,cufft,cufile,cupti,curand,cusolver,cusparse,nvjitlink,
nvrtc,nvtx]==13.0.2; platform_system == "Linux"->torch>=1.13.1->flair)
(12.6.3.3)
Requirement already satisfied: nvidia-nvjitlink==13.0.88.* in
/usr/local/lib/python3.13/dist-packages (from cuda-
toolkit[cublas,cudart,cufft,cufile,cupti,curand,cusolver,cusparse,nvjitlink,
nvrtc,nvtx]==13.0.2; platform_system == "Linux"->torch>=1.13.1->flair)
(13.0.88)
```

```
Requirement already satisfied: nvidia-cuda-nvrtc==13.0.88.* in
/usr/local/lib/python3.13/dist-packages (from cuda-
toolkit[cublas,cudart,cufft,cufile,cupti,curand,cusolver,cusparse,nvjitlink,
nvrtc,nvtx]==13.0.2; platform_system == "Linux"→torch≥1.13.1→flair)
(13.0.88)
Requirement already satisfied: nvidia-nvtx==13.0.85.* in
/usr/local/lib/python3.13/dist-packages (from cuda-
toolkit[cublas,cudart,cufft,cufile,cupti,curand,cusolver,cusparse,nvjitlink,
nvrtc,nvtx]==13.0.2; platform_system == "Linux"→torch≥1.13.1→flair)
(13.0.85)
Collecting huggingface-hub≥0.10.0 (from flair)
  Downloading huggingface_hub-0.36.2-py3-none-any.whl.metadata (15 kB)
Collecting tokenizers≤0.23.0,≥0.22.0 (from transformers<5.0.0,≥4.25.0
→transformers[sentencepiece]<5.0.0,≥4.25.0→flair)
  Downloading tokenizers-0.22.2-cp39-abi3-
manylinux_2_17_x86_64.manylinux2014_x86_64.whl.metadata (7.3 kB)
Requirement already satisfied: safetensors≥0.4.3 in
/usr/local/lib/python3.13/dist-packages (from transformers<5.0.0,≥4.25.0
→transformers[sentencepiece]<5.0.0,≥4.25.0→flair) (0.8.0)
Requirement already satisfied: sentencepiece≠0.1.92,≥0.1.91 in
/usr/local/lib/python3.13/dist-packages (from transformers[sentencepiece]
<5.0.0,≥4.25.0→flair) (0.2.2)
Requirement already satisfied: protobuf in /usr/local/lib/python3.13/dist-
packages (from transformers[sentencepiece]<5.0.0,≥4.25.0→flair) (6.33.6)
Requirement already satisfied: tenacity==9.1.4 in
/usr/local/lib/python3.13/dist-packages (from wikipedia-api≥0.5.7→flair)
(9.1.4)
Requirement already satisfied: anyio in /usr/local/lib/python3.13/dist-
packages (from httpx<1,≥0.23.0→huggingface-hub≥0.10.0→flair) (4.15.1)
Requirement already satisfied: certifi in /usr/local/lib/python3.13/dist-
packages (from httpx<1,≥0.23.0→huggingface-hub≥0.10.0→flair) (2026.7.22)
Requirement already satisfied: httpcore==1.* in
/usr/local/lib/python3.13/dist-packages (from httpx<1,≥0.23.0→huggingface-
hub≥0.10.0→flair) (1.0.9)
Requirement already satisfied: idna in /usr/local/lib/python3.13/dist-
packages (from httpx<1,≥0.23.0→huggingface-hub≥0.10.0→flair) (3.20)
Requirement already satisfied: h11≥0.16 in /usr/local/lib/python3.13/dist-
packages (from httpcore==1.*→httpx<1,≥0.23.0→huggingface-hub≥0.10.0
→flair) (0.16.0)
Requirement already satisfied: urllib3≠2.2.0,<3,≥1.25.4 in
/usr/local/lib/python3.13/dist-packages (from botocore<1.44.0,≥1.43.117
→boto3≥1.20.27→flair) (2.5.0)
Requirement already satisfied: cuda-pathfinder~=1.1 in
/usr/local/lib/python3.13/dist-packages (from cuda-bindings<14,≥13.0.3
→torch≥1.13.1→flair) (1.8.3)
Requirement already satisfied: cloudpickle≥3.0 in
/usr/local/lib/python3.13/dist-packages (from joblib≥1.2.0→scikit-
learn≥1.0.2→flair) (3.1.2)
Requirement already satisfied: attrs≥19.2.0 in
/usr/local/lib/python3.13/dist-packages (from jsonlines≥1.2.0
→bioc<3.0.0,≥2.0.0→flair) (26.1.0)
Requirement already satisfied: mpmath<1.4,≥1.1.0 in
/usr/local/lib/python3.13/dist-packages (from sympy≥1.13.3→torch≥1.13.1
→flair) (1.3.0)
```

```
Requirement already satisfied: accelerate⩾0.26.0 in
/usr/local/lib/python3.13/dist-packages (from
transformers[sentencepiece,torch]<5.0,⩾4.1→transformer-smaller-training-
vocab⩾0.2.3→flair) (1.15.0)
Requirement already satisfied: soupsieve>1.2 in
/usr/local/lib/python3.13/dist-packages (from beautifulsoup4→gdown⩾4.4.0
→flair) (3.0.3)
Requirement already satisfied: sortedcontainers in
/usr/local/lib/python3.13/dist-packages (from intervaltree
→bioc<3.0.0,⩾2.0.0→flair) (2.4.0)
Requirement already satisfied: MarkupSafe⩾2.0 in
/usr/local/lib/python3.13/dist-packages (from jinja2→mpld3⩾0.3→flair)
(3.0.4)
Requirement already satisfied: charset_normalizer<4,⩾2 in
/usr/local/lib/python3.13/dist-packages (from requests[socks]→gdown⩾4.4.0
→flair) (3.4.9)
Requirement already satisfied: PySocks≠1.5.7,⩾1.5.6 in
/usr/local/lib/python3.13/dist-packages (from requests[socks]→gdown⩾4.4.0
→flair) (1.7.1)
Requirement already satisfied: psutil in /usr/local/lib/python3.13/dist-
packages (from accelerate⩾0.26.0→transformers[sentencepiece,torch]
<5.0,⩾4.1→transformer-smaller-training-vocab⩾0.2.3→flair) (5.9.5)
Downloading flair-0.15.1-py3-none-any.whl (1.2 MB)
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 1.2/1.2 MB 58.5 MB/s eta 0:00:00
Downloading bioc-2.1-py3-none-any.whl (33 kB)
Downloading boto3-1.43.117-py3-none-any.whl (140 kB)
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 140.0/140.0 kB 17.0 MB/s eta
0:00:00
Downloading conllu-4.5.3-py2.py3-none-any.whl (16 kB)
Downloading deprecated-3.0.0-py3-none-any.whl (21 kB)
Downloading ftfy-6.3.1-py3-none-any.whl (44 kB)
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 44.8/44.8 kB 4.5 MB/s eta
0:00:00
Downloading mpld3-0.5.12-py3-none-any.whl (203 kB)
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 203.1/203.1 kB 23.3 MB/s eta
0:00:00
Downloading pytorch_revgrad-0.2.0-py3-none-any.whl (4.6 kB)
Downloading segtok-1.5.11-py3-none-any.whl (24 kB)
Downloading transformer_smaller_training_vocab-0.4.2-py3-none-any.whl (14
kB)
Downloading transformers-4.57.6-py3-none-any.whl (12.0 MB)
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 12.0/12.0 MB 118.1 MB/s eta
0:00:00
Downloading huggingface_hub-0.36.2-py3-none-any.whl (566 kB)
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 566.4/566.4 kB 47.8 MB/s eta
0:00:00
Downloading wikipedia_api-0.16.0-py3-none-any.whl (129 kB)
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 129.8/129.8 kB 14.3 MB/s eta
0:00:00
Downloading botocore-1.43.117-py3-none-any.whl (16.1 MB)
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 16.1/16.1 MB 109.8 MB/s eta
0:00:00
Downloading jmespath-1.1.0-py3-none-any.whl (20 kB)
Downloading jsonlines-4.0.0-py3-none-any.whl (8.7 kB)
```

```
Downloading s3transfer-0.19.3-py3-none-any.whl (90 kB)
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 90.3/90.3 kB 10.1 MB/s eta
0:00:00
Downloading tokenizers-0.22.2-cp39-abi3-
manylinux_2_17_x86_64.manylinux2014_x86_64.whl (3.3 MB)
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 3.3/3.3 MB 119.3 MB/s eta
0:00:00
Downloading intervaltree-3.2.1-py2.py3-none-any.whl (25 kB)
Building wheels for collected packages: langdetect, pptree, sqlitedict,
docopt
  Building wheel for langdetect (setup.py) ... done
  Created wheel for langdetect: filename=langdetect-1.0.9-py3-none-any.whl
size=993335
sha256=d8148a5d601c26235841311a18e9a50d93734a3ac3ed68dcd6d738acd5c2e6c3
  Stored in directory:
/root/.cache/pip/wheels/eb/87/25/2dddf1c94e1786054e25022ec5530bfed52bad86d88
2999c48
  Building wheel for pptree (setup.py) ... done
  Created wheel for pptree: filename=pptree-3.1-py3-none-any.whl size=4641
sha256=040e89761261d0d3c639cd8d9acf90b9e4fc98b131dc08d94f3cd2fc6b86678f
  Stored in directory:
/root/.cache/pip/wheels/97/5c/d1/a8347c9c986fc222c12240fafdde88e50a72553e19d
7517177
  Building wheel for sqlitedict (setup.py) ... done
  Created wheel for sqlitedict: filename=sqlitedict-2.1.0-py3-none-any.whl
size=16958
sha256=cf1316e356faf4f7fcb253dd5f590a23cdb8e9e74739bb3875e90d10a07e446c
  Stored in directory:
/root/.cache/pip/wheels/56/1e/68/fc282426da812281e81368e59ed19f3edc6d9eef451
8dbc41c
  Building wheel for docopt (setup.py) ... done
  Created wheel for docopt: filename=docopt-0.6.2-py2.py3-none-any.whl
size=13781
sha256=a87336a6cb8368c473379bdd7559408a55e41cb4c52da3656f717048d5b217ee
  Stored in directory:
/root/.cache/pip/wheels/0b/1d/03/175286677fb5a1341cc3e4755bf8ec0ed08f3329afd
67446b0
Successfully built langdetect pptree sqlitedict docopt
Installing collected packages: sqlitedict, pptree, docopt, segtok,
langdetect, jsonlines, jmespath, intervaltree, ftfy, deprecated, conllu,
huggingface-hub, botocore, bioc, wikipedia-api, tokenizers, s3transfer,
mpld3, transformers, boto3, pytorch-revgrad, transformer-smaller-training-
vocab, flair
  Attempting uninstall: huggingface-hub
    Found existing installation: huggingface_hub 1.33.0
    Uninstalling huggingface_hub-1.33.0:
      Successfully uninstalled huggingface_hub-1.33.0
  Attempting uninstall: tokenizers
    Found existing installation: tokenizers 0.23.3
    Uninstalling tokenizers-0.23.3:
      Successfully uninstalled tokenizers-0.23.3
  Attempting uninstall: transformers
    Found existing installation: transformers 5.20.1
    Uninstalling transformers-5.20.1:
```

```
      Successfully uninstalled transformers-5.20.1
ERROR: pip's dependency resolver does not currently take into account all
the packages that are installed. This behaviour is the source of the
following dependency conflicts.
diffusers 0.41.0 requires huggingface-hub<2.0,⩾1.32.0, but you have
huggingface-hub 0.36.2 which is incompatible.
gradio 6.30.0 requires huggingface-hub<3.0,⩾1.16.0, but you have
huggingface-hub 0.36.2 which is incompatible.
gradio-client 2.7.2 requires huggingface-hub<3.0,⩾1.16.0, but you have
huggingface-hub 0.36.2 which is incompatible.
Successfully installed bioc-2.1 boto3-1.43.117 botocore-1.43.117 conllu-
4.5.3 deprecated-3.0.0 docopt-0.6.2 flair-0.15.1 ftfy-6.3.1 huggingface-hub-
0.36.2 intervaltree-3.2.1 jmespath-1.1.0 jsonlines-4.0.0 langdetect-1.0.9
mpld3-0.5.12 pptree-3.1 pytorch-revgrad-0.2.0 s3transfer-0.19.3 segtok-
1.5.11 sqlitedict-2.1.0 tokenizers-0.22.2 transformer-smaller-training-
vocab-0.4.2 transformers-4.57.6 wikipedia-api-0.16.0
```

In [2]:

```
import torch
import torch.nn as nn
import torch.optim as optim
from torch.utils.data import Dataset, DataLoader
import numpy as np
import random

from flair.data import Sentence
from flair.embeddings import FlairEmbeddings
```

In [3]:

```
SEED = 42
random.seed(SEED)
torch.manual_seed(SEED)
```

Out[3]:

```
<torch._C.Generator at 0x7a7076813690>
```

Tworzymy słownik

In [4]:
```
sentences = [
  "It was a cold morning in October.",
  "Anna stood by the window, watching the fog rise over the city.",
  "She wrapped her scarf tighter and checked her phone.",
  "A message from Tom appeared: 'Are you ready for today?'.",
  "She smiled, remembering their plan.",
  "They were going to meet at the train station at dawn.",
  "The streets were silent, except for the sound of her footsteps.",
  "As she turned the corner, she saw a figure waiting in the mist.",
  "It was him."
]

# Inicjalizacja embeddingów
embedding = FlairEmbeddings('news-forward')

# dla każdego zdania tworzymy obiekty Sentence
flair_sentences = [Sentence(s) for s in sentences]

# Wydobycie embeddingów słów
for s in flair_sentences:
    embedding.embed(s)

# Zbudowanie słownika słów na podstawie korpusu
words = set(" ".join(sentences).split())
word2idx = {word: i+1 for i, word in enumerate(words)}  # indeksy od 1 (0 = padding)
idx2word = {i: word for word, i in word2idx.items()}
vocab_size = len(word2idx) + 1

print("Słownik:", word2idx)
print("Rozmiar słownika:", vocab_size)
```

```
2026-10-22 13:14:59,371 https://flair.informatik.hu-berlin.de/resources/embeddings/flair/news-forward-0.4.1.pt not found in cache, downloading to /tmp/tmpcjkn6501
```

```
100%|██████████| 69.7M/69.7M [00:04<00:00, 14.7MB/s]
```

```
2026-10-22 13:15:04,832 copying /tmp/tmpcjkn6501 to cache at /root/.flair/embeddings/news-forward-0.4.1.pt
2026-10-22 13:15:04,876 removing temp file /tmp/tmpcjkn6501
```

```
Słownik: {'As': 1, 'except': 2, 'she': 3, 'city.': 4, 'you': 5, 'window,': 6, "'Are": 7, 'Tom': 8, 'fog': 9, 'were': 10, 'from': 11, 'over': 12, 'streets': 13, "today?'.": 14, 'going': 15, 'to': 16, 'smiled,': 17, 'in': 18, 'scarf': 19, 'sound': 20, 'morning': 21, 'Anna': 22, 'wrapped': 23, 'him.': 24, 'cold': 25, 'stood': 26, 'for': 27, 'The': 28, 'her': 29, 'figure': 30, 'turned': 31, 'train': 32, 'plan.': 33, 'their': 34, 'station': 35, 'dawn.': 36, 'message': 37, 'saw': 38, 'ready': 39, 'was':
```

```
40, 'the': 41, 'October.': 42, 'by': 43, 'rise': 44, 'appeared:': 45, 'A':
46, 'remembering': 47, 'silent,': 48, 'a': 49, 'She': 50, 'of': 51,
'corner,': 52, 'tighter': 53, 'They': 54, 'footsteps.': 55, 'at': 56,
'checked': 57, 'mist.': 58, 'meet': 59, 'It': 60, 'and': 61, 'watching': 62,
'phone.': 63, 'waiting': 64}
Rozmiar słownika: 65
```

Każde słowo dostaje unikalny numer.

Indeks 0 zarezerwowany na padding (wyrównywanie długości sekwencji).

Przewidywanie kolejnego słowa

Będzi trenować lstm w kontekście całej sekwencji, żeby przewidywał nam kolejne słowo. Np: "To był chłodny" → "poranek"

Tworzymy dataset na podstawie nazszych zdań

In [5]:
```
import torch
import torch.nn as nn
import torch.optim as optim


all_words = " ".join(sentences).split() # wszystkie słowa (to co mamy w
słowniku)
seq_length = 4 # będziemy przewidywać piąte słowo na podstawie 4 poprzednich


X_data, y_data = [], []
# sztucznie tworzymy przykłady
for i in range(len(all_words) - seq_length):
  # przesuwamy nasze okienko sekwencji słów po tekście
  X_seq = all_words[i:i+seq_length]
  y_seq = all_words[i+seq_length]
  # nie używamy słów tylko index ze slownika
  X_data.append([word2idx[w] for w in X_seq])
  y_data.append(word2idx[y_seq])


# robimy tensory na których będziemy trenować
X = torch.tensor(X_data)
y = torch.tensor(y_data)


print(f"Sample input: {X_data[0]} → target: {y_data[0]}")
```

```
Sample input: [60, 40, 49, 25] → target: 21
```

Na co patrzymy?

Przygotowaliśmy sobie mechanizm że model dostaje 4 słowa (o indeksach 45, 20, 13, 17), a jego zadaniem jest przewidzieć kolejne słowo (o indeksie 5).

Stwórzmy model

Stwórzmy model który od razu zrobi to co ręcznie zrobiliśmy u góry

In [6]:
```
class WordPredictorLSTM(nn.Module):
  def __init__(self, vocab_size, embedding_dim, hidden_dim):
    super().__init__()
    # to nasz embeding modelu - go też będziemy trenować!
    # każde słowo zamieniamy na wektor o długości embedding_dim
    self.embedding = nn.Embedding(vocab_size, embedding_dim)
    # tu jest nasz model :)
    self.lstm = nn.LSTM(embedding_dim, hidden_dim, batch_first=True)
    # mapujemy wynik LSTM na rozkład prawdopodobieństwa z wszystkich słów w
słowniku
    self.fc = nn.Linear(hidden_dim, vocab_size)


  def forward(self, x):
    # x - przyszło do nas słowo (index w słowniku)
    # zmieniamy index na wektor
    embeds = self.embedding(x)
    # przepuszczamy przez lstm
    # out: ukryte stany dla każdego kroku czasowego,
    # h: ostatni ukryty stan (zawiera podsumowanie sekwencji!) - Ważne !,
    # c: wewnętrzny stan pamięci.
    out, (h, c) = self.lstm(embeds)
    # ostatni ukryty stan, zwracamy prawdopodobieństwa - które słowo ma
najwyzsze prawdopodobieństwo?
    return self.fc(h[-1])


model_word = WordPredictorLSTM(vocab_size=vocab_size, embedding_dim=100,
hidden_dim=128)
criterion = nn.CrossEntropyLoss()
optimizer = optim.Adam(model_word.parameters(), lr=0.001)
```

Trening

In [7]:
```
EPOCHS = 200
for epoch in range(EPOCHS):
  optimizer.zero_grad() # czyścimy gradienty
  output = model_word(X) # przewidywania modelu
  loss = criterion(output, y) # porównujemy przewidywanie vs rzeczywiste
słowo
  loss.backward() # liczymy gradienty
  optimizer.step() # aktualizujemy wagi
  if (epoch+1) % 40 == 0:
    print(f"Epoch {epoch+1}/{EPOCHS}, Loss: {loss.item():.4f}")
```

```
Epoch 40/200, Loss: 1.5260
Epoch 80/200, Loss: 0.0797
Epoch 120/200, Loss: 0.0263
Epoch 160/200, Loss: 0.0151
Epoch 200/200, Loss: 0.0100
```

Przetestujmy nasz model! 😺

In [8]:
```
def generate_text(model, seed_text, next_words=8):
  model.eval() # nie aktualizuj wag
  words = seed_text.split() # pierwsze zdanie zamieniamy na listę słów
  for _ in range(next_words): # generujemy tyle kolejnych słów ile chcemy
    seq = [word2idx.get(w, 0) for w in words[-seq_length:]] # zmieniamy na
indeksy nasz kontekst, seq_length zdefiniowaliśmy podczas tworzenia datasetu
    x = torch.tensor(seq).unsqueeze(0) # tensor o wymiarze (1, długość
kontekstu)
    with torch.no_grad():
      output = model(x) # przewidujemy na podstawie otrzymanych 4 słów
      predicted_idx = torch.argmax(output, dim=1).item() # wybieramy jakie
słowo ma największe prawdopodobieństwo
      predicted_word = idx2word.get(predicted_idx, '') # indext wybranego
słowa → słowo
      words.append(predicted_word)
  return ' '.join(words) # zwracamy wszystko po kolei (nasza historia)


seed = "It was a cold"
print("Generated continuation:")
print(generate_text(model_word, seed))
```

```
Generated continuation:
It was a cold morning in October. Anna stood by the window,
```

Zadanie

3 pkt

Stwórz sieć LSTM, która przewidzi kolejne zdanie a nie kolejne słowo

Input sentence → Encoder LSTM → Hidden state → Decoder LSTM → Predicted next sentence

UWAGA! Zproponowano szkielet LSTM SeqToSeq - ale jeśli chcesz zrobić to w inny sposób, użyć innych emebedingów itp - śmiało :)

In [20]:
```
sentences = [
    "It was a bright and sunny morning.",
    "The birds chirped happily in the trees.",
    "A cool breeze swept through the streets.",
    "Children played joyfully in the park.",
    "The cat lounged lazily on the windowsill.",
    "The coffee shop smelled of freshly brewed espresso.",
    "Raindrops pattered softly against the roof.",
    "The sky turned shades of pink and orange at sunset.",
    "A dog barked loudly at a passing car.",
    "The leaves rustled under the feet of walkers.",
    "Smoke rose from the chimney into the crisp air.",
    "The waves crashed rhythmically onto the shore.",
    "A lone cyclist rode along the quiet road.",
    "Flowers bloomed vibrantly in the garden.",
    "The old bookstore had a comforting aroma of paper.",
    "Fog rolled over the hills in the early morning.",
    "A gentle hum of traffic filled the city streets.",
    "The clock struck noon with a resounding chime.",
    "Windows rattled in the wind of the approaching storm.",
    "The bakery displayed an array of fresh pastries.",
    "A distant train whistle echoed through the valley.",
    "Stars twinkled faintly in the darkening sky.",
    "The playground was empty as dusk approached.",
    "A fisherman cast his line into the calm river.",
    "The scent of pine trees lingered in the air.",
    "Clouds gathered ominously over the mountains.",
    "The market buzzed with people and chatter.",
    "A child laughed as a balloon floated away.",
    "The lamp post flickered in the evening gloom.",
    "Shadows stretched long across the empty street.",
    "The fountain gurgled peacefully in the square.",
    "A paper airplane sailed across the classroom.",
    "The dog chased its tail in playful circles.",
    "The train station was crowded with travelers.",
    "Snowflakes drifted silently from the gray sky.",
    "A painter worked diligently on his canvas.",
    "The wind whispered secrets through the trees.",
    "A street musician played a soulful tune.",
    "The river sparkled under the midday sun.",
    "The fire crackled warmly in the hearth.",
    "A cyclist narrowly avoided a puddle on the road.",
    "The library was silent except for turning pages.",
    "The airplane soared high above the clouds.",
    "A butterfly landed delicately on a flower.",
    "The aroma of freshly baked bread filled the kitchen.",
    "The old clock ticked steadily in the hallway.",
```

```
    "A cat chased a shadow across the wall.",
    "The garden gate creaked in the evening breeze.",
    "The moon reflected off the still lake.",
    "A boy flew a kite on the open field.",
    "The streets glistened after a sudden rain shower."
]
```

In [21]:

```
# EMBEDINGI
import flair
from flair.embeddings import DocumentPoolEmbeddings
import torch.nn.functional as F

flair.device = torch.device('cpu')
doc_embedding = DocumentPoolEmbeddings([FlairEmbeddings('news-forward')])
flair_sentences = [Sentence(s) for s in sentences]
doc_embedding.embed(flair_sentences)

# jeśli tu jest zero to prawdopodobnie coś poszło nie tak :)
print("Embedding pierwszego zdania:", flair_sentences[0].embedding.shape)

# DATASET - sequence to sequence
all_emb = torch.stack([s.embedding for s in flair_sentences]).detach()
X_seq = all_emb[:-1].unsqueeze(1)
y_seq = all_emb[1:].unsqueeze(1)

# jeśli tu jest zero to prawdopodobnie coś poszło nie tak :)
print("X_seq wymiar:", X_seq.shape)

# MODEL
class Seq2SeqLSTM(nn.Module):
    def __init__(self, input_dim, hidden_dim):
        super().__init__()
        self.encoder = nn.LSTM(input_dim, hidden_dim, batch_first=True)
        self.decoder = nn.LSTM(hidden_dim, hidden_dim, batch_first=True)
        self.fc = nn.Linear(hidden_dim, input_dim)

    def forward(self, x):
        # encoder:
        _, (h, c) = self.encoder(x)
        # decoder:
        dec_in = h[-1].unsqueeze(1)
        out, _ = self.decoder(dec_in, (h, c))

        return self.fc(out)

input_dim = X_seq.shape[2]
hidden_dim = 64
model_seq = Seq2SeqLSTM(input_dim, hidden_dim)


optimizer = optim.Adam(model_seq.parameters(), lr=0.005)
EPOCHS = 500
```

```
for epoch in range(EPOCHS):
    model_seq.train()
    optimizer.zero_grad()
    output = model_seq(X_seq)
    loss = 1 - F.cosine_similarity(output, y_seq, dim=-1).mean()
    loss.backward()
    optimizer.step()
    if (epoch+1) % 50 == 0:
        print(f"Epoch {epoch+1}/{EPOCHS}, Loss: {loss.item():.4f}")
```

```
Embedding pierwszego zdania: torch.Size([2048])
X_seq wymiar: torch.Size([50, 1, 2048])
Epoch 50/500, Loss: 0.1858
Epoch 100/500, Loss: 0.1483
Epoch 150/500, Loss: 0.0719
Epoch 200/500, Loss: 0.0391
Epoch 250/500, Loss: 0.0233
Epoch 300/500, Loss: 0.0140
Epoch 350/500, Loss: 0.0078
Epoch 400/500, Loss: 0.0045
Epoch 450/500, Loss: 0.0024
Epoch 500/500, Loss: 0.0012
```

UWAGA: **Twój model nie jest modelem generatywnym**! Twój LSTM przewiduje embedding następnego zdania, a my znajdziemy w zbiorze zdań to, które jest najbardziej podobne.

In [22]:
```
import torch
import torch.nn.functional as F

# Podajemy zdanie startowe (ostatnie zdanie z naszej sekwencji)

X_input =
torch.tensor(flair_sentences[-1].embedding.cpu().detach().numpy()).unsqueeze
(0).unsqueeze(0)

# Model w trybie ewaluacyjnym
model_seq.eval()
with torch.no_grad():
    predicted_embedding = model_seq(X_input)

print("Wymiary przewidzianego embedingu", predicted_embedding.shape)

# Zamieniamy na 1D tensor
pred_emb = predicted_embedding.squeeze(0).squeeze(0)
# dużo problemów w AI to niezgadzające się shape'y - PAMIĘTAJ PRINTOWAĆ
ROZMIAR jesli coś nie gra

# Tworzymy listę embeddingów wszystkich zdań w zbiorze
emb_list = [s.embedding for s in flair_sentences]
# Obliczamy podobieństwo cosinusowe do wszystkich zdań
similarities = [F.cosine_similarity(pred_emb, e, dim=0) for e in emb_list]
# Wybieramy indeks najbardziej podobnego zdania
best_idx = torch.argmax(torch.tensor(similarities))
# wygląda jak niedawne zadanie z labów z wektorami słów, prawda? :)

print('Początek historii: ', flair_sentences[-1])
print("Przewidziane kolejne zdanie (najbliżej w naszym zestawie):")
print(sentences[best_idx])
```

```
Wymiary przewidzianego embedingu torch.Size([1, 1, 2048])
Początek historii:  Sentence[9]: "The streets glistened after a sudden rain
shower."
Przewidziane kolejne zdanie (najbliżej w naszym zestawie):
A cat chased a shadow across the wall.
```